

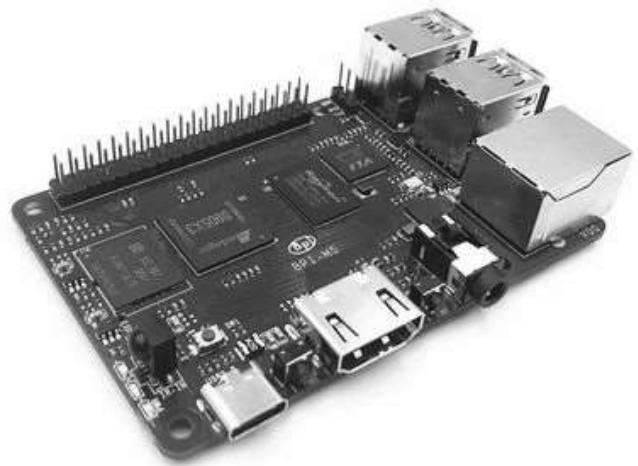
Banana Pi BPI-M5

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BPI-M5 Introduction

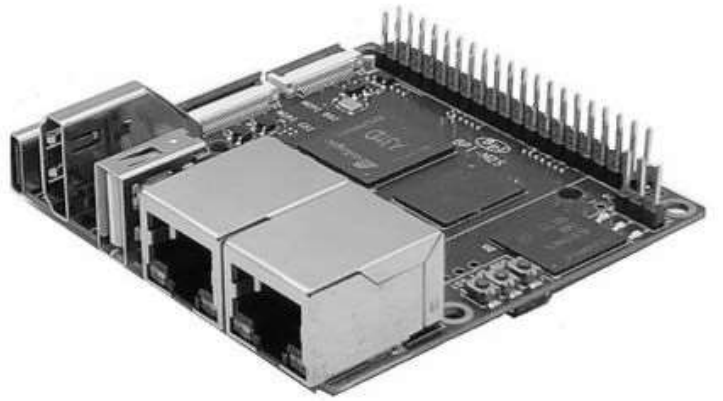
Banana Pi BPI-M5 is the next generation single board computer from Banana Pi in 2020. It is powered by Amlogic S905X3 quad-core Cortex-A55 (2.0 XXGHz) processor. Onboard 4GB LPDDR4 memory and 16GB EMMC storage, and supports 4 USB 3.0 interface, a gigabit network port.



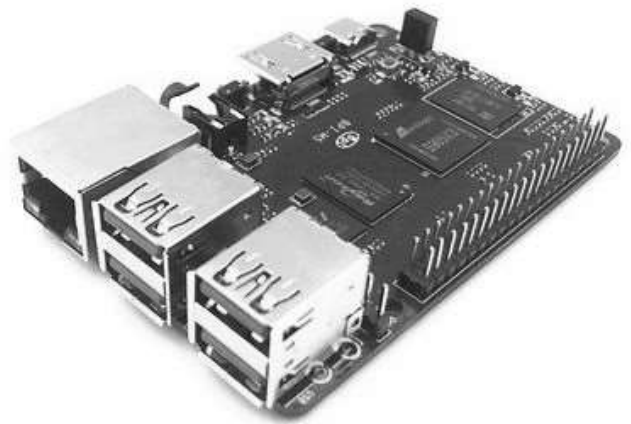
Banana Pi BPI-M5



Banana Pi BPI-M2 Pro S905x3 design



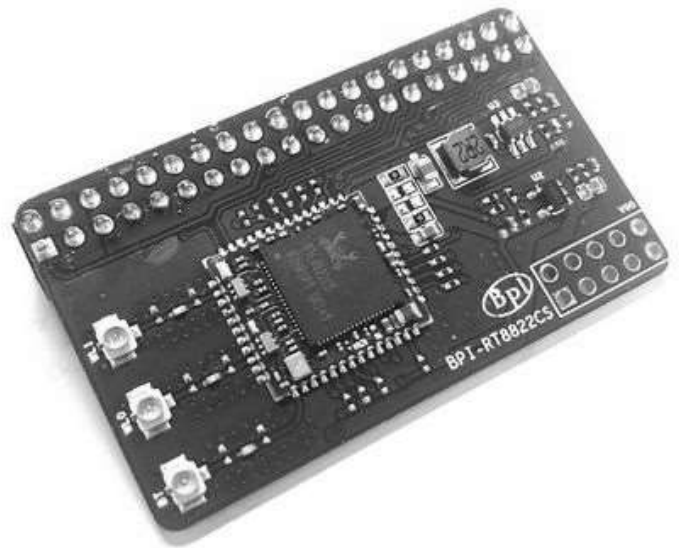
Banana Pi BPI-M2S Amlogic A311D/S922 chip



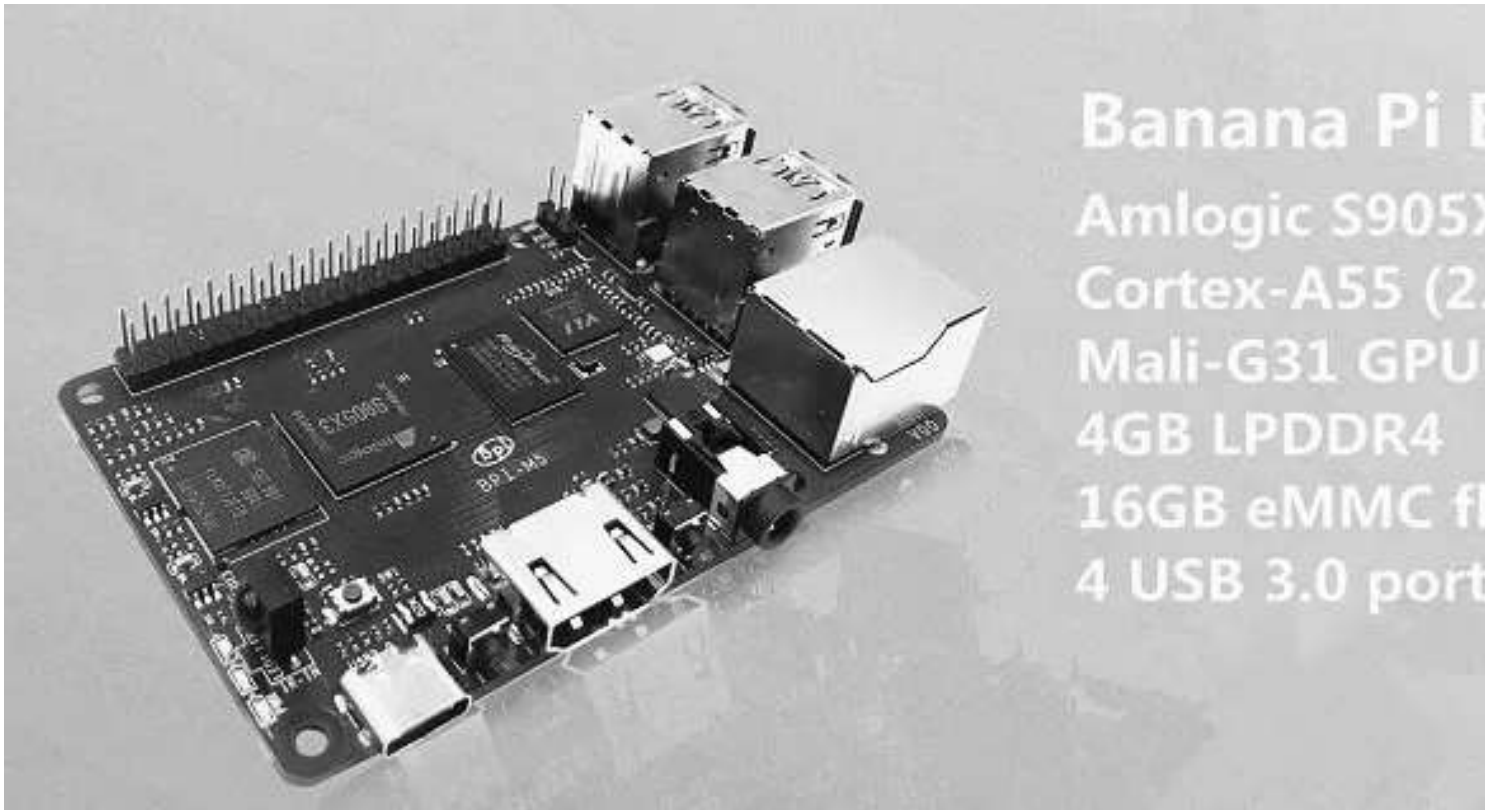
Amlogic S905X3 Processor



BPI-M5 case design



WiFi&BT board, 802.11 a/b/g/n/ac 2T2R WiFi and Bluetooth 5.0



Banana Pi M5
Amlogic S905X3
Cortex-A55 (2.0GHz)
Mali-G31 GPU
4GB LPDDR4
16GB eMMC flash
4 USB 3.0 ports

Key Features

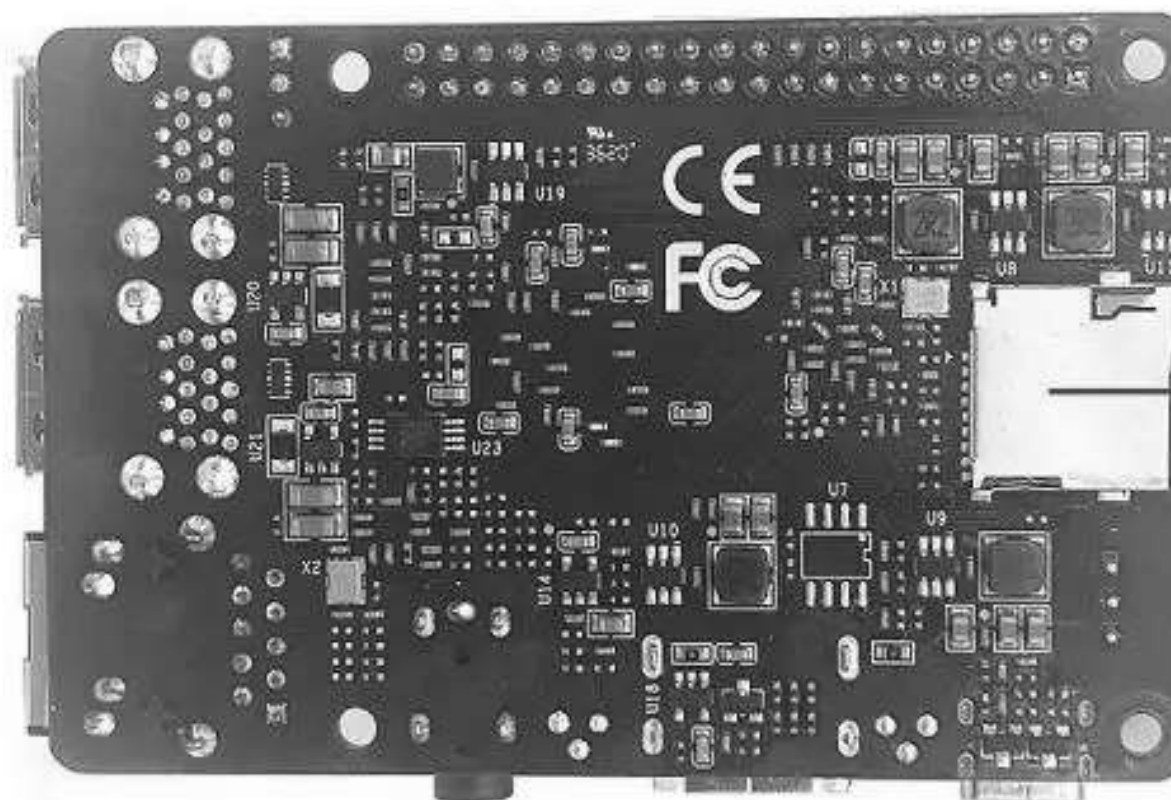
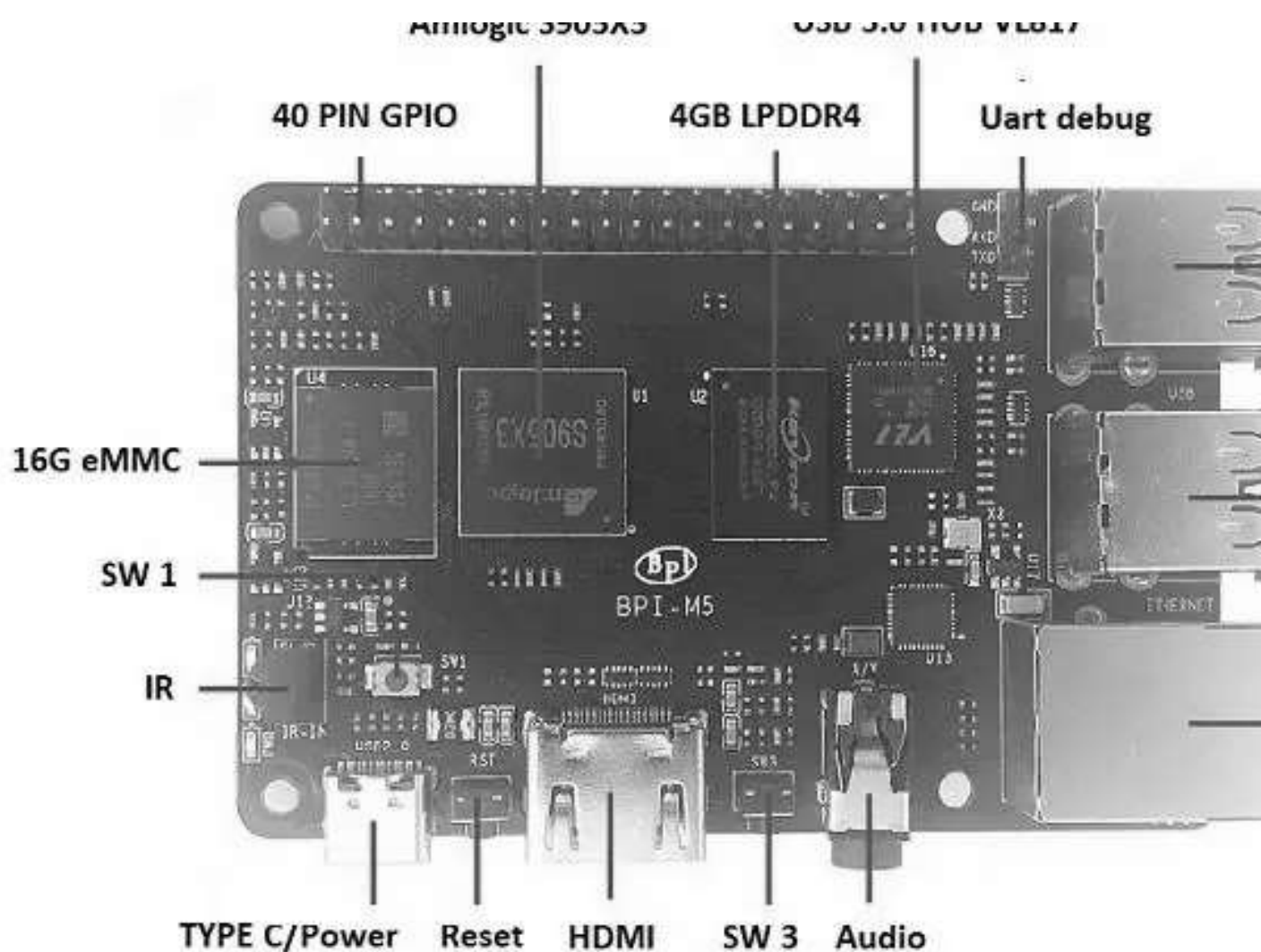
- Amlogic S905X3 quad core Cortex-A55 (2.0xxGHz)
- Mali-G31 GPU
- 4GB LPDDR4
- 16GB eMMC flash
- 4 USB 3.0
- 1 GbE ethernet
- HDMI output

Getting Start

- Getting Started with BPI-M5

Hardware

Hardware interface



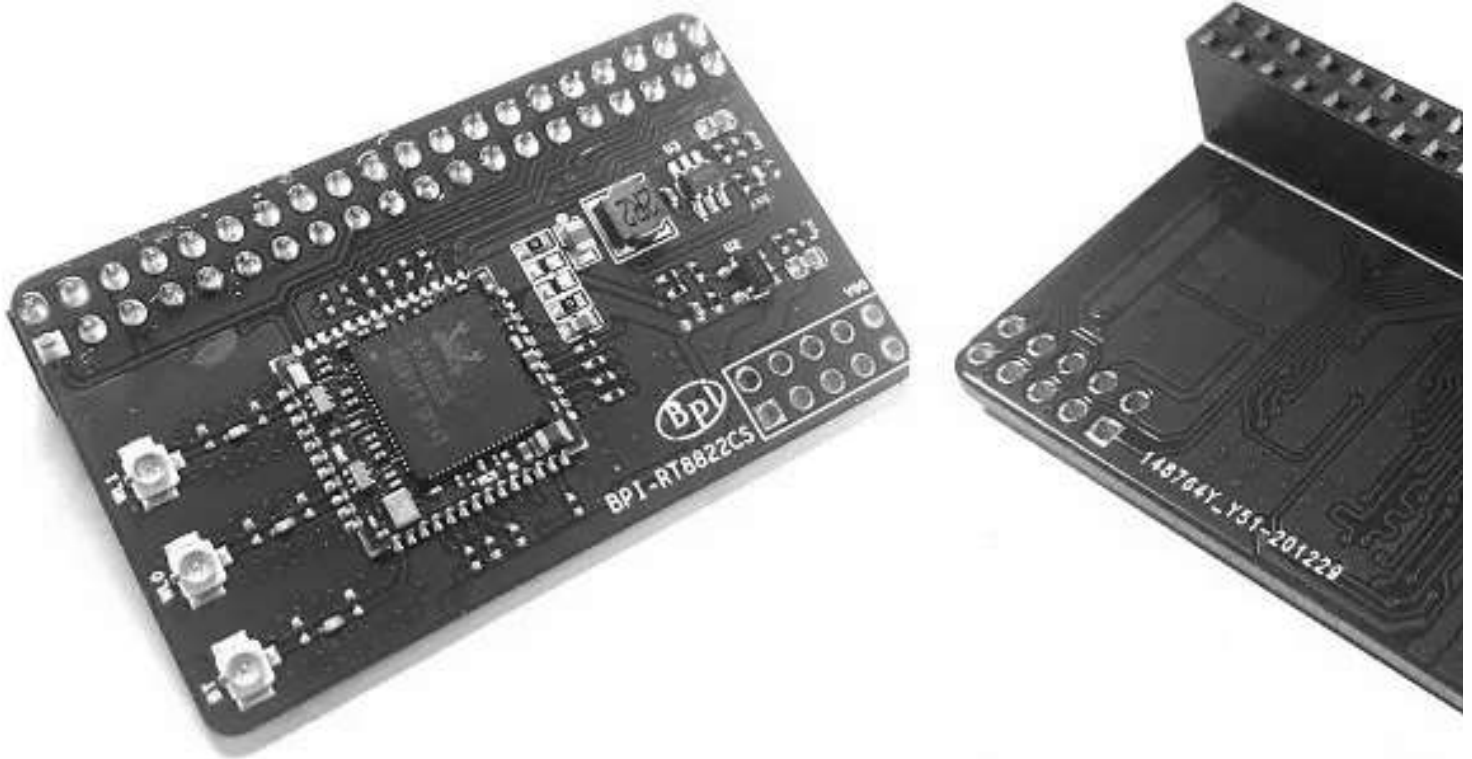
Hardware spec

HardWare Specification of Banana pi BPI-M5	
CPU	Amlogic S905X3 Quad-Core Cortex-A55 (2.0xxGHz)
GPU	Mali-G31 MP2 GPU with 4 x Execution Engines (650Mhz)
Memory	4 GB LPDDR4
Storage	MicroSD slot with support for up to 256GB expansion and 16G eMMC flash with support for up to 64GB
Network	10/100/1000 Mbit/s Ethernet ,Optional WiFi USB dongle
Video Output(s)	1 x HDMI 2.0 (up to 4K@60Hz with HDR, CEC, EDID)
Audio Output(s)	3.5mm jack and 1 x HDMI digital output
USB ports	USB 3.0 PORT (x4)
GPIO	40 Pin Header : GPIO (x28) and Power (+5V, +3.3V and GND). GPIO pins can be used for UART, I2C, SPI or PWM
Switches	Reset, Power and U-boot
LED	Power Status and Activity status
Power Source	5 volt @3A via Micro USB (TYPE C)
Size & Weight	92x60mm, 48g
OS	Android and Linux

Wifi & BT support via expansion board

SDIO interface Wifi&BT

- WiFi&BT board, 802.11 a/b/g/n/ac 2T2R WiFi and Bluetooth 5.0 , support BPI-M5 and BPI-F2P

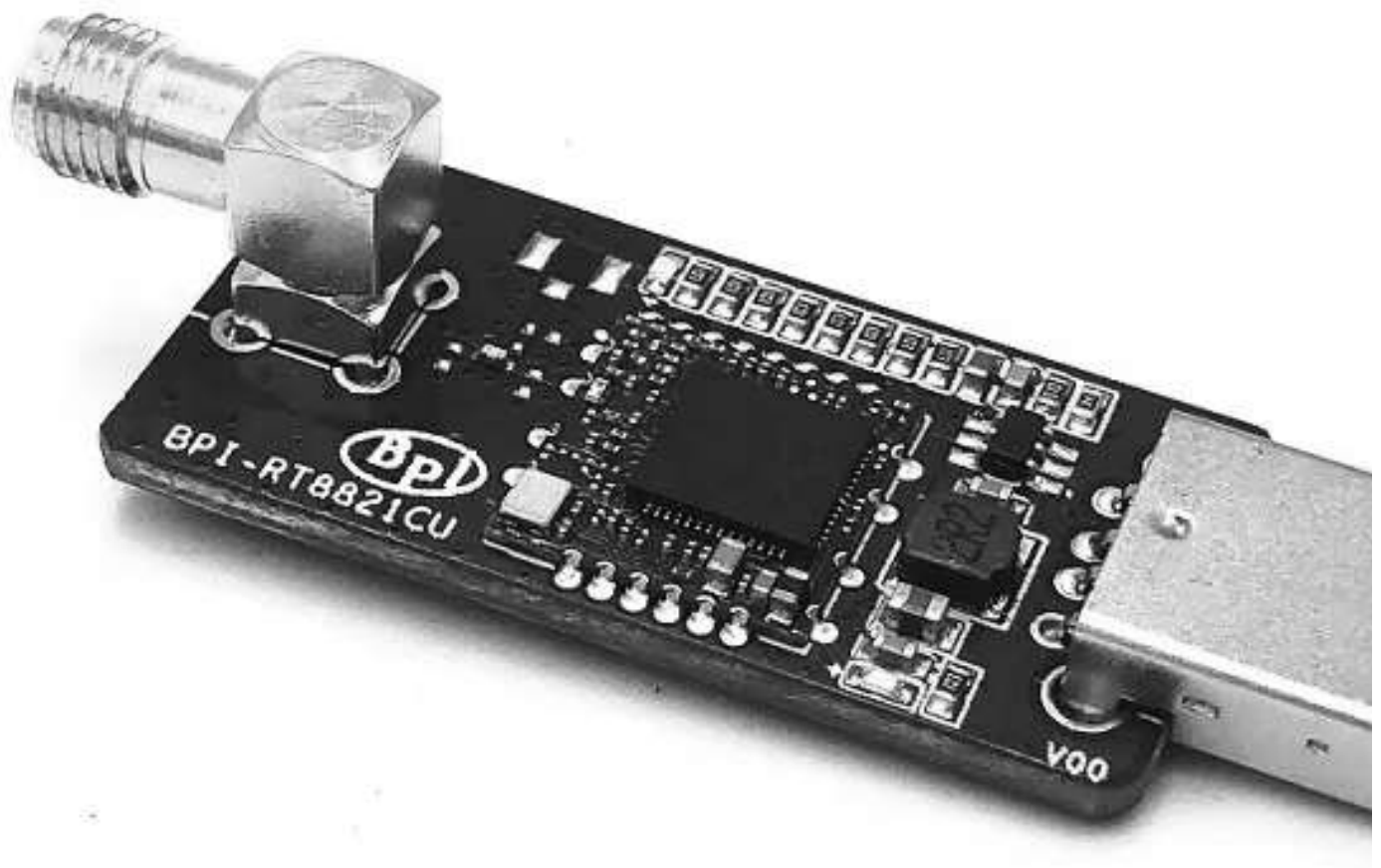


802.11 a/b/g/n/ac 2T2R WiFi and Bluetooth 5.0

- how to use : http://wiki.banana-pi.org/Getting_Started_with_BPI-M5#Wifi.2FT_support
- discuss on forum : <http://forum.banana-pi.org/t/bpi-m5-wifi-bt-board-sdio-interface-802-11-a-b-g-n-ac-2t2r-wifi-and-bluetooch-5-0/11846>
- easy to buy sample : <https://www.aliexpress.com/item/1005002550783568.html?spm=5261.ProductManageOnline.0.0.34ed4edfgdV59j>

standard USB interface Wifi&BT

- Banana Pi Wifi&BT 4.2 expansion Board, standard USB interface, so support all open source boards via USB port.IEEE 802.11b/g/n/ac(1T1R) USB WLANAnd BT Module



- how to use : http://wiki.banana-pi.org/Getting_Started_with_BPI-M5#Wifi.2FBT_support
- discuss on forum : <http://forum.banana-pi.org/t/banana-pi-wifi-bt-4-2-expansion-board-standard-usb-interface/12162>

BPI-M2 Pro VS BPI-M5 VS Odroid C4 VS RPI 4

BPI-M2 Pro VS BPI-M5 VS Odroid C4 VS RPI 4				
	Banana Pi BPI-M2 Pro	Banana Pi BPI-M5	Odroid C4	Raspberry Pi 4
Processor	Amlogic S905X3 Quad-Core Cortex-A55 ARMv8.2-A 64-bit 1.5GHz	Amlogic S905X3 Quad-Core Cortex-A55 ARMv8.2-A 64-bit 1.5GHz	Amlogic S905X3 Quad-Core Cortex-A55 ARMv8.2-A 64-bit 1.5GHz	Broadcom BCM2711, Quad core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5GHz
GPU	Mali-G31 MP2 GPU	Mali-G31 MP2 GPU	Mali-G31 MP2 GPU	Broadcom VideoCore VI
RAM	2GB LPDDR4	4GB LPDDR4	DDR4 4GiB with 32-bit bus width	2GB, 4GB or 8GB LPDDR4-3200 SDRAM
Storage	micro SD card slot & 16GB eMMC flash on board	micro SD card slot & 16GB eMMC flash on board	micro SD card slot , eMMC connector for option	micro SD card slot, non EMMC
Network	1 GbE ,WiFi&BT5.0 onbard	GbE ,Optional WiFi USB dongle	GbE	GbE, 2.4 GHz and 5.0 GHz IEEE 802.11ac wireless, Bluetooth 5.0, BLE
Video Output	1 x HDMI 2.1 (up to 4K@60Hz with HDR, CEC, EDID)	1 x HDMI 2.1 (up to 4K@60Hz with HDR, CEC, EDID)	1 x HDMI 2.1 (up to 4K@60Hz with HDR, CEC, EDID)	2 × micro-HDMI 2.0
Audio Output	1 x HDMI digital output	3.5mm jack and 1 x HDMI digital output	Optional SPDIF optical output	3.5 mm analogue audio-video jack
DSI	N/A	N/A	N/A	2-lane MIPI DSI display port
CSI	N/A	N/A	N/A	2-lane MIPI CSI camera port
USB	USB 3.0 x 2 1 Micro USB 2.0	USB 3.0 x 4 TYPE C	USB 3.0 x 4, 1 x Micro USB 2.0 port (OTG)	2 USB 3.0 ports; 2 USB 2.0 ports;
Power	5 volt @3A via DC power	5 volt @3A via Micro USB (TYPE C)	1 x DC power jack (Outer diameter : 5.5mm,	5V DC via USB-C connector (minimum 3A*)

			inner diameter : 2.1mm)		
IR	Yes	Yes	Yes	N/A	
GPIO	40 Pin Header : GPIO (x28) and Power (+5V, +3.3V and GND).	40 Pin Header : GPIO (x28) and Power (+5V, +3.3V and GND).	40 x GPIO pins	40-pin GPIO header, populated	
Button	Reset, user and U- boot	Reset, user and U- boot	N/A	N/A	
LED	Power Status and Activity status	Power Status and Activity status	Power Status and Activity status	Power Status and Activity status	
Dimensions & weight	65mm x 65mm,58g	85mm x 56mm, 48g	85mm x 56mm 59g	88 mm × 58 mm, 46 g	
System Support	Linux & Android	Linux & Android	Linux & Android	Linux & Android	

GPIO PIN define

BPI-M5 40PIN GPIO (CON2)

GPIO Pin Name	Default Function	Function2	Function3
CON1-P01	VCC-3V3		
CON1-P02	VCC5V		
CON1-P03	GPIOX_17	I2C_M2_SDA	
CON1-P04	VCC5V		
CON1-P05	GPIOX_18	I2C_M2_SCL	
CON1-P06	GND		
CON1-P07	GPIOX_5 SDIO_CMD		
CON1-P08	GPIOX_12	UART_A_TX	
CON1-P09	GND		
CON1-P10	GPIOX_13	UART_A_RX	
CON1-P11	GPIOX_3 SDIO_D3		
CON1-P12	GPIOAO_8	TDMB_SCLK	
CON1-P13	GPIOX_4 SDIO_CLK		
CON1-P14	GND		
CON1-P15	GPIOX_7 PWM_F		
CON1-P16	GPIOX_0 SDIO_D0		
CON1-P17	VCC-3V3		
CON1-P18	GPIOX_1 SDIO_D1		
CON1-P19	GPIOX_8 PCM_DIN		
CON1-P20	GND		
CON1-P21	GPIOX_9 PCM_DOUT		
CON1-P22	GPIOX_2 SDIO_D2		
CON1-P23	GPIOX_11	PCM_CLK	
CON1-P24	GPIOX_10	PCM_SYNC	
CON1-P25	GND		
CON1-P26	GPIOX_16	PWM_E	
CON1-P27	GPIOA_14	I2C_M3_SDA	

CON1-P28	GPIOA_15	I2C_M3_SCL	
CON1-P29	GPIOX_14	UART_A_CTS	
CON1-P30	GND		
CON1-P31	GPIOX_15	UART_A_RTS	
CON1-P32	GPIOX_19	PWM_B	
CON1-P33	GPIOX_6 PWM_A		
CON1-P34	GND		
CON1-P35	GPIOAO_7	TDMB_FS	
CON1-P36	GPIOH_5 SPDIF_IN		
CON1-P37	GPIOAO_7	I2S_MCLK	
CON1-P38	GPIOAO_10	TDMB_DIN	SPDIF_OUT
CON1-P39	GND		
CON1-P40	GPIOAO_4	TDMB_DOUT	

BPI-M5 Debug UART(CON1)

CON2-P1	GND
CON2-P2	UART0-RX
CON2-P3	UART0-TX

Development

Source code

Android

android 9 source code: <https://github.com/BPI-SINOVOIP/BPI-S905X3-Android9>

How to build the Android Source Code (https://wiki.banana-pi.org/Getting_Started_with_BPI-M5#Build_Android_Source_Code)

Linux BSP source code

Linux BSP source code: <https://github.com/BPI-SINOVOIP/BPI-M5-bsp>

How to build the Linux BSP Source Code (https://wiki.banana-pi.org/Getting_Started_with_BPI-M5#Build_Linux_Source_Code)

Resources

- Because of the Google security update some of the old links will not work if the images you want to use cannot be downloaded from the new link bpi-image Files (https://drive.google.com/drive/folders/0B_YnvHgh2rwjVjNyS2pheEtWQlk?resourcekey=0-U4TI84zIBdId7bHHjf2qKA)
- All banana pi docement(SCH file,DXF file,and doc) (<https://drive.google.com/drive/folders/0B4PAo2nW2Kfndjh6SW9MS2xKSWs?resourcekey=0-qXGFXXKmd7AVy0S81OXM1RA&usp=sharing>)
- BPI-M5 schematic diagram (<https://download.banana-pi.dev/d/3ebbfa04265d4dddb81b/files/?p=%2FDocuments%2FBPI-M5%2FBPI-M5-SCH-V10-Release.pdf>)
- BPI-M5 PCB DXF file (<https://download.banana-pi.dev/d/3ebbfa04265d4dddb81b/files/?p=%2FDocuments%2FBPI-M5%2FBPI-M5-PCB-V10-DXF.rar>)
- Amlogic S905x3 datasheet (https://download.banana-pi.dev/d/3ebbfa04265d4dddb81b/files/?p=%2FDocuments%2FBPI-M5%2FS905X3_Public_Datasheet_Hardkernel.pdf)
- Banana Pi BPI-M5 CE,FCC,RoHS Certification (https://wiki.banana-pi.org/Product_certification)
- Install OpenGapps on Bananapi BPI-M5 Android 9.0:<https://www.youtube.com/watch?v=fXOKmWfpqF8>

System image

Android

How to flash Android Image (https://wiki.banana-pi.org/Getting_Started_with_BPI-M5#Android)

- 2021-05-20 release, Android 9.0 with kernel 4.9.113, mbox variant ui image

Download: 2021-05-20-bpi-m5-m2pro-mbox-android9-emmc.zip (<https://download.banana-pi.dev/d/ca025d76afd448aabc63/files/?p=%2FImages%2FBPI-M5%2Fandroid9%2F2021-05-20-bpi-m5-m2pro-mbox-android9-emmc.zip>)
MD5: dda75a075a5d7572756f57e4a256dffa

- 2021-05-20 release, Android 9.0 with kernel 4.9.113, tablet variant ui image

Download: 2021-05-20-bpi-m5-m2pro-tablet-android9-emmc.zip (<https://download.banana-pi.dev/d/ca025d76afd448aabc63/files/?p=%2FImages%2FBPI-M5%2Fandroid9%2F2021-05-20-bpi-m5-m2pro-tablet-android9-emmc.zip>)
MD5: 27381d5e307034e5c5456b167f67ef75

Linux

How to flash Linux Image (https://wiki.banana-pi.org/Getting_Started_with_BPI-M5#Linux)

Ubuntu

- 2021-06-21 release, Ubuntu Mate Desktop 20.04 with kernel 4.9.241

Download: 2021-06-21-ubuntu-20.04-mate-desktop-bpi-m5-m2pro-aarch64-sd-emmc.img.zip (<https://download.banana-pi.dev/d/ca025d76afd448aabc63/files/?p=%2FImages%2FBPI-M5%2Fubuntu%2F2021-06-21-ubuntu-20.04-mate-desktop-bpi-m5-m2pro-aarch64-sd-emmc.img.zip>)
MD5: cd6bc3cb4fba1b83e2e3f63fd49ba15b

- 2021-06-21 release, Ubuntu Server 20.04 with kernel 4.9.241

Download: 2021-06-21-ubuntu-20.04-server-bpi-m5-m2pro-aarch64-sd-emmc.img.zip (<https://download.banana-pi.dev/d/ca025d76afd448aabc63/files/?p=%2FImages%2FBPI-M5%2Fubuntu%2F2021-06-21-ubuntu-20.04-server-bpi-m5-m2pro-aarch64-sd-emmc.img.zip>)
MD5: 5c896c37eaa590cfbb8a6c7a5ba433a0

Debian

- 2021-06-21 release, Debian Buster with kernel 4.9.241

Download: 2021-06-21-debian-10-buster-bpi-m5-m2pro-aarch64-sd-emmc.img.zip (<https://download.banana-pi.dev/d/ca025d76afd448aabc63/files/?p=%2FImages%2FBPI-M5%2Fdebian%2F2021-06-21-debian-10-buster-bpi-m5-m2pro-aarch64-sd-emmc.img.zip>)
MD5: 4ea1d71a5ba1a8fa743e4c8abbe7569b

Third part image

CoreELEC

- CoreELEC 19.2

CoreELEC (<https://coreelec.org/>) official support Bananapi M5 and M2Pro since 19.2-Matrix_rc1

Download: <https://coreelec.org/#download>

Install Guide: <https://coreelec.org/#install>

Source Code: <https://github.com/CoreELEC/CoreELEC>

how to install: <https://wiki.coreelec.org/coreelec:bpim2pro>

Armbian

Armbian unstable legacy and current build for Bananapi M5 and M2Pro

Google Drive:

https://drive.google.com/drive/folders/1Cx_DA90Tbr8_M_8eQd-NwFUOp3UTwhYt

Baidu Cloud : <https://pan.baidu.com/s/15y4LN4K-w0vrwMTcg5VGQ>

Pincode: j9aj

Volumio

Download:

Google drive:

<https://drive.google.com/drive/folders/1B7nsy4Jxt2lBgXoA6XEnaamEQvZyF>

Baidu cloud : <https://pan.baidu.com/s/1mj9OC8P2VSI5GAMnIKGBqA>

PinCode: mrrb

User Guide: <https://cdn.volumio.org/wp-content/uploads/2019/01/Quick-Start-Guide-Volumio.pdf>

Source code: <https://github.com/Dangku/volumio-build>

Platform prebuild package: <https://github.com/Dangku/volumio-platform-bananapi>

Development guide:

https://volumio.github.io/docs/User_Manual/Quick_Start_Guide.html

Odroid Android image

Base on Odroid android source code and build for bananapi m5/m2pro

Download:

Google Drive:

https://drive.google.com/drive/folders/1u_CddlHvpAMu2VZJfIfUKTsFFyigC

Install Guide: <https://forum.odroid.com/viewtopic.php?f=204&t=38579>

Ubuntu Core 20

Ubuntu Core 20 demo image for bananapi m5/m2pro.

Download:

Google drive: <https://drive.google.com/drive/folders/1TdAQ-HdggKrOPmWi0chpHDJY0SMWIUaC>

Baidu cloud :

Install Guide: <https://ubuntu.com/core/docs/uc20/install>

Archlinux

- Archlinux xfce and minimal demo image, kernel 4.9, image build refer to Archdroid (<https://archdroid.org/>)

login: alarm/alarm, or root/root

Download:

Google

drive:<https://drive.google.com/drive/folders/1rSSNgbseY2mwTMlIauGR1yrXl>

Baidu link : https://pan.baidu.com/s/1Ck_H51jPUyAv98o73I8oAw Pincode: rbv2

discuss on forum : <https://forum.banana-pi.org/t/bpi-m5-bpi-m2-pro-new-image-archlinux-2021-9-14/12595>

HuaWei OpenEuler

- OpenEuler demo image, bpi kernel 4.9

Download:

Baidu link : <https://pan.baidu.com/s/12b7q3y-m3YRyD7GwhKE0QA>

Pincode: lv50

SIG gitee link : <https://gitee.com/openeuler/raspberrypi>

Discuss on fourm : <http://forum.banana-pi.org/>

Manjaro for BPI-M5

Manjaro porting for BPI-M5 from Official Image for Odroid C4, the ported image use kernel 5.10. If anyone is interested on give it a try, the download link is below.

- Download Link: <https://mega.nz/folder/BuZWkLhC#phC9KSsassB4bmkAegYyKA>
- discuss on forum : <https://forum.banana-pi.org/t/manjaro-for-bpi-m5/12726>

Easy to buy

- buy it form aliexpress (https://pt.aliexpress.com/item/1005001871842695.html?spm=a2g0o.productlist.0.0.59976fff0GL35q&algo_pvid=24dfc96b-f6c1-4850-9ff4-fd177ee36648&algo_expid=24dfc96b-f6c1-4850-9ff4-fd177ee36648-0&btsid=0bb47a2216147586025791703e886c&ws_ab_test=searchweb0_0,searchweb201602_,searchweb201603_)
- easy to buy RTL8822 wifi module sample :
<https://www.aliexpress.com/item/1005002550783568.html?spm=5261.ProductManageOnline.0.0.34ed4edfgdV59j>
- Mass order,please contact : judyhuang@banana-pi.com

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